Expander Graphs in Cryptography

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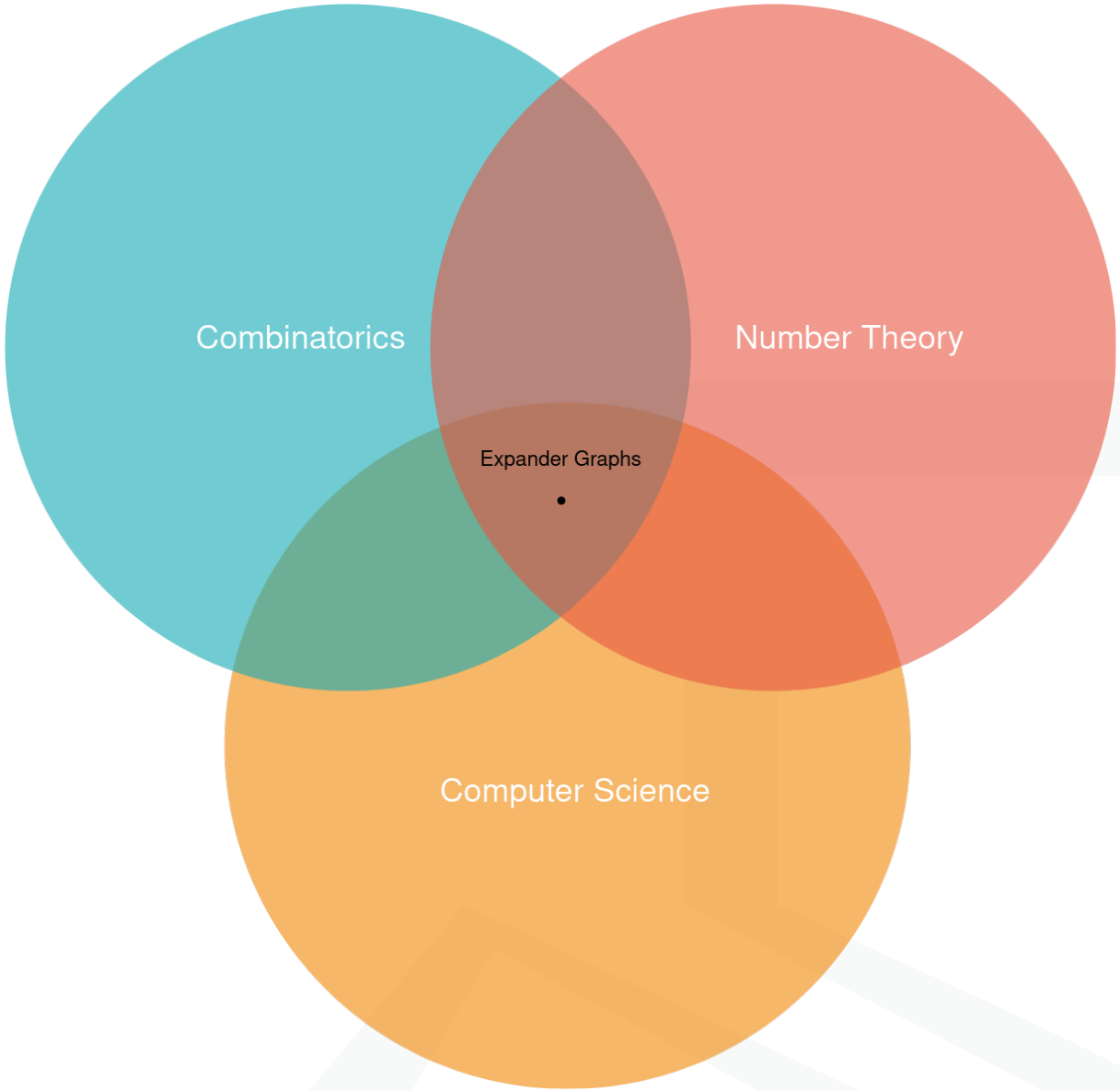
Expander Graphs



- **Part 1:** Motivation and Definitions
- **Part 2:** Existence and Explicit Construction
- **Part 3:** Expander Graphs in Cryptography

Part 1: Motivation and Definitions

Some Applications

- Designing efficient error correcting codes - **use an expander graph.**
- Complexity analysis (i.e. $SL = L$) - **expander graphs.**
- Pseudorandom number generation - **expander graphs.**
- Reliable networks - **expander graphs.**
- Cryptographic hash functions - **expander graphs!**
- Riemann Hypothesis - **expander graphs?**

Two Important Properties

On a 'good' expander graph:

- Path finding is hard.
- Random walks converge to the uniform distribution.

Expander Graphs - Three Ways

We are interested in connected, regular (multi)-graphs. Expander graphs may be described in three languages:

- **Combinatorial:** no small vertex cuts, high connectivity.
- **Algebraic:** small second eigenvalue of adjacency matrix.
- **Probabilistic:** random walks converge quickly.

In fact, these properties are equivalent!

Boundary of a Graph

Let $G = (V, E)$ be a d-regular graph (all vertices have degree d).
Let $S \subseteq V$. The *boundary* of S is

$$\delta S = \{(u, v) \in E : u \in S, v \in V - S\}$$

From now on we will refer to d-regular graphs on n vertices as $[n, d]$-graphs.

Combinatorial Definition: Edge Expansion Ratio

Definition (Expansion Ratio)

The (edge) expansion ratio of G, $h(G)$ is defined as:

$$h(G) = \min_{\{S \subseteq V, |S| \leq \frac{|V|}{2}\}} \frac{|\delta S|}{|S|}$$

G is an expander if $h(G) > 0$.

- All connected graphs are expanders.
- Larger $h(G)$ means a better expander.
- Simplest description but hardest to compute.

Graph Spectral Analysis

- The *adjacency matrix* A of a graph on n vertices is an $n \times n$ matrix where each $a_{i,j} = \#$ edges from v_i to v_j.
- A is symmetric and real $\implies$ n eigenvalues.

Definition (Graph Spectrum)

The *spectrum* of a graph is the set of eigenvalues of A where:

$$d \geq \lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq -d$$

Some useful properties of graph spectrum:

- If G is d-regular, then $\lambda_1 = d$.
- G is bipartite if.f $\lambda_1 = -\lambda_n$.
- G is connected if.f $\lambda_1 > \lambda_2$.

Algebraic Definition: Spectral Gap

Given an $[n, d]$-graph G with spectrum $d = \lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n \geq -1$:

- *Spectral gap* of G is defined as $d - \lambda_2$.
- Related to expansion ratio by the following bound:
$$\frac{1}{2}(d - \lambda_2) \geq h(G) \geq \sqrt{2d(d - \lambda_2)}$$
- G is an $[n, d, \alpha]$-graph if $\lambda_2 \leq \alpha$. Expander if $\alpha < d$.
- An infinite family $\{G_i\}$ of $[n_i, d, \alpha]$-graphs is an *expander family* if for all i, $\alpha < d$.
- Another important parameter: $\lambda(G) = max(\{|\lambda_2|, ..., |\lambda_n|\})$.

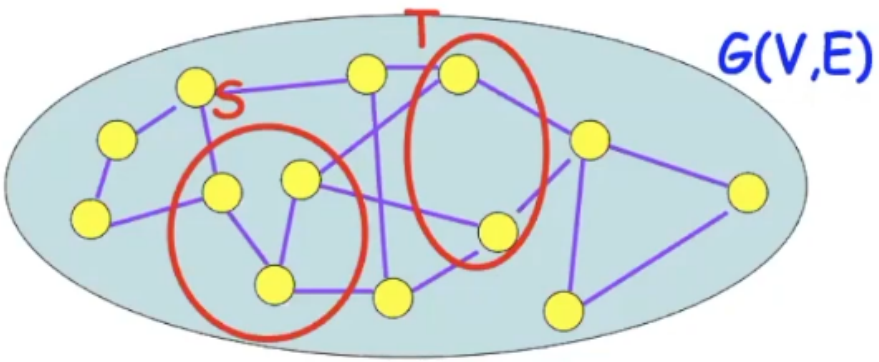
Expander Mixing Lemma

Theorem

Let $G = (V, E)$ be an $[n, d]$-graph. Then for all $S, T \subseteq V$:

$$|E(S,T)| - \frac{d|S||T|}{n} \leq \lambda(G)\sqrt{|S||T|}$$

- $|E(S,T)| = \#$ edges between vertices in S and T.
- In loose terms: the number of edges between any two large vertex sets is about what you expect in a random $[n, d]$-graph.
- Corollary: random walks on the graph converge to uniform distribution in $O(log(n))$ steps.



How good can we get?

Theorem (Alon-Boppana)

If G is an $[n, d]$-graph, then

$$\lambda(G) \geq 2\sqrt{d-1} - \epsilon.$$

for $\epsilon > 0$ where $\epsilon \to 0$ as $n \to \infty$.

- Smaller $\lambda(G)$ means a better expander.

Definition (Ramanujan graphs)

An $[n, d]$-graph is *Ramanujan* if

$$\lambda(G) \leq 2\sqrt{d-1}$$

- Ramanujan graphs are the best possible expanders.

Part 2: Existence, Explicit Construction and Open Problems in Expanders

Existence of sparse expanders

Theorem (Pinsker 1973)

Almost every 3-regular graph is an expander.

Proof.

Exercise. (Hint: Probabilistic method.) □

Challenge is explicit construction of low degree expander families.

Constructing Expander Graphs

Two types of construction. Given an infinite family $\{G_i\}$ of $[n_i, d]$-expander-graphs:

- **Weakly Explicit**: G_i can be constructed in polynomial time. (Polynomial in n_i).
- **Strongly Explicit**: Given any integer i, a vertex $v \in V(G_i)$ and a $k \in \{1, ..., d\}$, the k-th neighbour of v can be computed in polynomial time. (Polynomial in length of input (i, v, k)).
- Typically constructed algebraically. i.e. cayley graphs

Example: A Strongly Explicit Family of Expanders

A family of 3-regular graphs $G_p = (V_p, E_p)$ for every prime p:

- $V_p = \mathbb{Z}_p$
- Each vertex x is connected to $x + 1, x - 1, x^{-1} \pmod{p}$.
- $(0, 0) \in E_p$.
- For every p, this graph is strongly explicit.

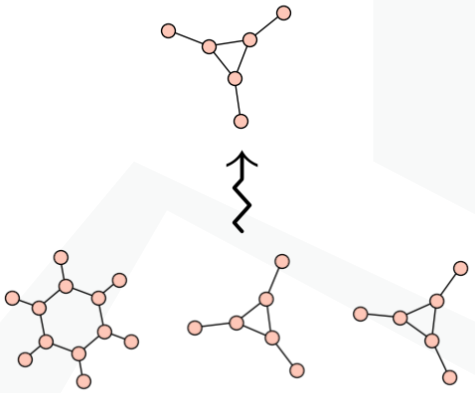
Open Problem in Ramanujan graphs

Problem

Does there exist infinite families of d-regular non-bipartite Ramanujan graphs for each $d \geq 3$?

Two (constructive) approaches to this problem:

- **Cayley graphs:** start with a group and a generating set, get a graph.
- **Covering graphs or lifts:** Analogous to topological covering spaces - build a bigger graph from a smaller graph.
- **Question!**



Solutions

- **LPS Graphs (Lubotzsky, Phillips, Sarnak 1988):** solved for $d = q + 1$ where q is a prime power. Using Cayley graphs of PSL_2 and PGL_2 over finite fields.
- **Supersingular Isogeny Graphs (Pizer 1990):** proven to be Ramanujan, degree $p + 1$.
- Other constructions (zig-zag product, interlacing polynomials, lifts of Ramanujan graphs). Bipartite case proven for all d in 2013.
- Problem remains open for non-bipartite cases where $d - 1$ is not a prime power. i.e. $d = 7$

Expanders: What we know so far

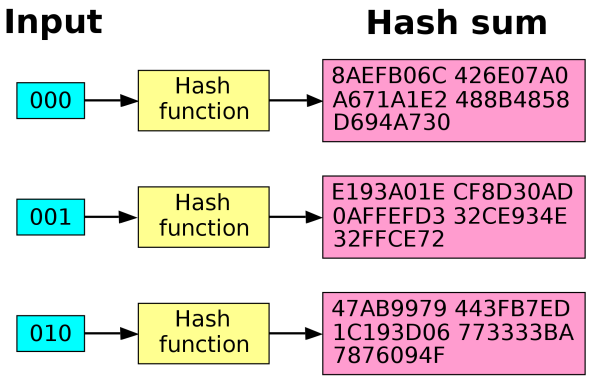
- Ramanujan graphs are optimal expanders.
- Good expanders are sparse and finding paths on them is hard.
- Can we use them in cryptography?

Part 3: Expander Graphs in Cryptography

4 Important Properties of Cryptographic Hash Functions

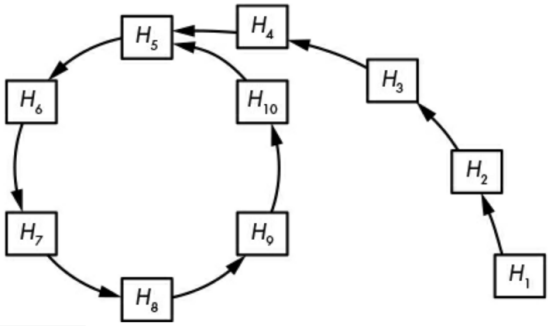
Given a hash function h:

- **Efficiency:** computing the hash of an input is fast.
- **Pre-image resistance**: given an image y, it is hard to find an input x such that $h(x) = y$.
- **Second pre-image resistance**: Given an input x_1, finding an input x_2 such that $h(x_1) = h(x_2)$ is hard.
- **Collision resistance**: It is hard to find two inputs x_1, x_2 such that $h(x_1) = h(x_2)$.



Hash Functions using Expander Graphs

- Input used as directions for a walk on the graph (no backtracking).
- Output: end vertex of a walk.
- Two candidate expander families proposed: LPS and Supersingular Isogeny Graphs(Charles, Goren, Lauter 2006).
- Security assumption: non-trivial cycles yield collisions.
- LPS based hash function highly efficient.
- Unfortunately neither collision nor pre-image resistant.
- Lauter-Petit Attack (2008) finds collisions and preimages probabilistically in polynomial time.
- Isogeny based variants exist which are secure, but slow.



State of Modern Public Key Cryptography

- 'One way' functions - encrypt with public key, hard to decrypt without private key.
- Public key ciphers used today: RSA, DH, ECDH.
- Cryptographic primitives - discrete log and integer factorisation.
- Shor's algorithm (1994) - quantum computers can decrypt in polynomial time.
- Search for post-quantum cryptographic primitives. (NIST draft standards available 2022-2024).
- Isogeny based cryptography a leading candidate (closely related to expander graphs)

Elliptic Curves and Isogenies

- **Elliptic Curves**.
  $y^2 = x^3 + ax + b$
  Algebraic and geometric structure. Set of solutions over a field form a group.
- Morphisms of elliptic curves are called **isogenies**.
  Isogenies preserve group & geometric structure.

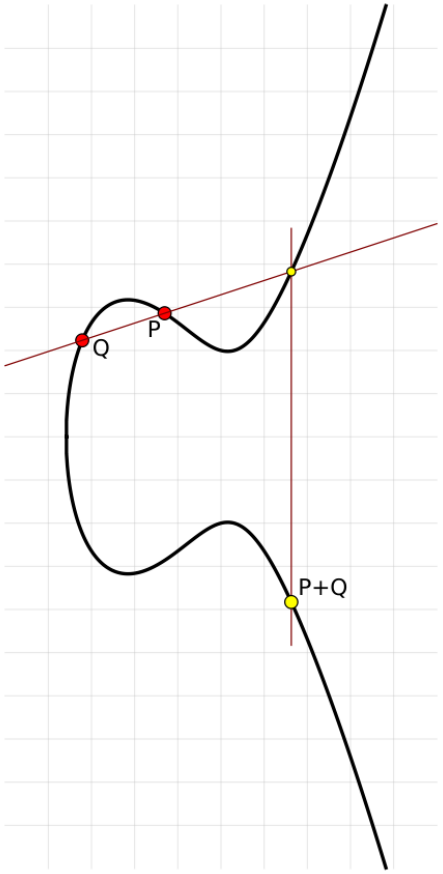


Figure 1: Group operation on $y^2 = x^3 - 4x + 7$

ℓ-Isogeny Graphs

- Pick your favourite field K, and prime ℓ.
- Define a graph: **vertices** are all the elliptic curves over K, **edges** are isogenies between them of fixed degree ℓ.
- This graph has rich structure. Two classes of connected components.
- One **supersingular** component. The rest are called **ordinary** and form 'volcanoes' - finite if K is finite.
- Supersingular component is Ramanujan.

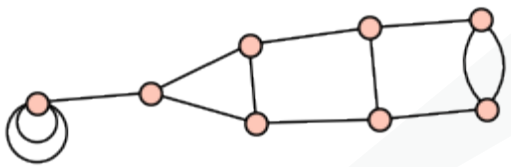


Figure 2: Supersingular component $K = \mathbb{F}_{97^2}$, $\ell = 2$

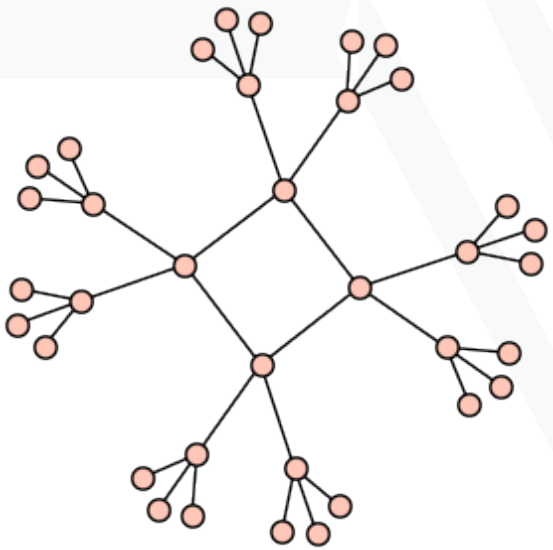


Figure 3: A volcano graph

The Hard Problem of Isogeny Based Cryptography

- **Isogeny Path Finding Problem:** Given the supersingular ℓ-isogeny graph over a finite field, and two elliptic curves, find a path between them.
- At most as hard as path finding on a Ramanujan graph.
- But could be easier?

Supersingular Isogeny Diffie Hellman (SIDH)

- Supersingular Isogeny Based Cryptography (SIDH) a leading candidate in this category.
- Based on a commutative action similar to DH.
- Partial image of isogenies given.
- Based on a more constrained security assumption.

$$\begin{tikzcd} E \arrow[r, "\phi"] \arrow[d, "\psi"] & E/\langle A \rangle \arrow[d, "\psi'"] \\ E/\langle B \rangle \arrow[r, "\phi'"] & E/\langle A, B \rangle \end{tikzcd}$$

Conclusion

- Leading post-quantum algorithms rely on the hardness assumption of path-finding on expander graphs.
- Constructing hash functions using Ramanujan graphs has been a subject of great research, but so far impractical.
- Questions?